

Perfect! I've analyzed the project structure. Now let me create a comprehensive implementation plan.

## Implementation Plan: Adding "Airline" Domain

### Project Analysis Summary

The project is a **LangChain-based Test Case Generator** with a clean separation of concerns:

- **Frontend:** React + TypeScript (Vite)
- **Backend:** Node.js + TypeScript + Express
- **Architecture:** Domain-specific prompts, RAG-based test case generation using LLMs

Currently supports 3 domains: **Banking, Insurance, Hospital**

---

### Step-by-Step Implementation Plan

#### STEP 1: Update Domain Types & Constants

##### Files to modify:

1. **frontend/src/services/api.ts**
2. **backend/src/factories/promptFactory.ts**

##### Changes:

- Add "airline" to the [Domain](#) type union
  - Add airline option to [DOMAINS](#) array with label "Airline"
  - Update existing domain-validation logic
- 

#### STEP 2: Create Airline-Specific Prompt

##### File to modify:

- **backend/src/factories/promptFactory.ts**

##### Changes:

- Create airlinePrompt ChatPromptTemplate with the provided system prompt
  - Add airline domain terminology (PNR, E-Ticket, GDS, IATA, Fare Class, etc.)
  - Add to [promptsByDomain](#) mapping dictionary
-

**STEP 3: Update Validation Schema**

**File to modify:**

- `backend/src/chains/testCaseChain.ts`

**Changes:**

- Update Zod schema enum to include "airline" in domain validation
- 

**STEP 4: Update API Route**

**File to modify:**

- `backend/src/routes/testCaseRoute.ts`

**Changes:**

- Ensure it imports updated DOMAINS (already generic, may need no changes)
- 

**Summary Table: Files & Changes**

| File                                   | Type           | Changes Required                                                           |
|----------------------------------------|----------------|----------------------------------------------------------------------------|
| frontend/src/services/api.ts           | Type/Config    | Add "airline" to Domain type, DOMAINS array                                |
| backend/src/factories/promptFactory.ts | Prompt Factory | Add "airline" to Domain type, create airlinePrompt, add to promptsByDomain |
| backend/src/chains/testCaseChain.ts    | Schema         | Add "airline" to Zod enum validation                                       |
| backend/src/routes/testCaseRoute.ts    | Route Handler  | No changes needed (generic)                                                |

---

**Expected Outcome**

- ✓ Airline option appears in Frontend domain selector dropdown
- ✓ Backend validates "airline" domain requests
- ✓ Airline-specific prompt with proper terminology (PNR, E-Ticket, GDS, IATA, etc.) is used

- ✓ Test cases generated with airline domain context
- ✓ No existing functionality is affected

After implementation –



